

1. IT Help Desk

Case Study

ABC Institute has 2,000 students and 150 staff members. Users frequently experience password issues, software installation requests, printer problems, and internet connectivity issues. The management plans to establish an IT Help Desk to improve support services.

Questions

1. Define an IT Help Desk and explain its main objectives.
2. Differentiate between IT Help Desk and Service Desk.
3. Identify five common responsibilities of a Help Desk officer.
4. Explain the ticketing process used in a Help Desk environment.
5. Analyze the problems faced by ABC Institute without a Help Desk.
6. Propose a workflow for handling student password reset requests.
7. Evaluate the benefits of implementing a centralized Help Desk.
8. Design a KPI framework to measure Help Desk performance.
9. Recommend suitable software tools for the Help Desk.
10. Create an improvement plan for ABC Institute's Help Desk operations.

Viva Questions

1. What is a Help Desk?
2. What is a support ticket?
3. What is ticket escalation?
4. What are Help Desk KPIs?
5. Why is customer satisfaction important?
6. What is first-call resolution?
7. Explain ticket prioritization.
8. What skills should a Help Desk technician possess?
9. What software is commonly used for Help Desk management?
10. How would you handle an angry user?

2. Incident Management

Case Study

The LMS system of a university becomes unavailable during final examinations. Thousands of students cannot access exam materials.

Questions

1. Define Incident Management.
2. Explain the Incident Management lifecycle.
3. Differentiate between Incident and Problem.
4. Identify stakeholders involved in managing this incident.
5. Analyze the impact of the LMS outage.
6. Determine the priority level of the incident and justify.
7. Evaluate the effectiveness of escalation procedures.
8. Recommend corrective actions.
9. Design an incident response workflow.
10. Create an incident report for management.

Viva Questions

1. What is an incident?
2. What is incident prioritization?
3. What is escalation?
4. Difference between incident and problem?
5. What is incident closure?
6. What is severity?
7. What is impact?
8. Why is communication important during incidents?
9. What is a major incident?
10. What metrics are used in Incident Management?

3. Service Level Agreement (SLA)

Case Study

XYZ Software Company provides technical support to a bank under a service contract guaranteeing 99.9% system availability.

Questions

1. Define SLA.
2. Explain the components of an SLA.
3. Differentiate SLA, OLA, and UC.
4. Identify SLA metrics in the case study.
5. Analyze the risks of not meeting SLA targets.
6. Evaluate the effectiveness of a 99.9% availability target.
7. Recommend SLA monitoring methods.
8. Design an SLA dashboard.
9. Develop SLA clauses for support services.
10. Create an SLA review plan.

Viva Questions

1. What is SLA?
2. What is uptime?
3. What is downtime?
4. What is service availability?
5. What are SLA penalties?
6. What is response time?
7. What is resolution time?
8. Why are SLAs important?
9. What is an SLA breach?
10. How do organizations monitor SLAs?

4. ITIL Framework

Case Study

A university IT department wants to improve service quality and adopt international best practices.

Questions

1. Define ITIL.
2. Explain the purpose of ITIL.
3. Describe key ITIL practices.
4. Identify ITIL processes relevant to the university.
5. Analyze benefits of ITIL implementation.
6. Evaluate challenges of adopting ITIL.
7. Recommend suitable ITIL practices.
8. Design an ITIL-based service workflow.
9. Develop an ITIL implementation roadmap.
10. Create a continuous improvement strategy.

Viva Questions

1. What is ITIL?
2. What is a service?
3. What is continual improvement?
4. What is Incident Management?
5. What is Change Management?
6. What is Problem Management?
7. What is Service Value System?
8. Why is ITIL popular?
9. What is a service catalog?
10. What are ITIL practices?

5. Customer Relationship Management (CRM)

Case Study

A private educational institute struggles to maintain communication with students and parents.

Questions

1. Define CRM.
2. Explain CRM objectives.
3. Identify CRM features.
4. Analyze customer communication issues.
5. Evaluate CRM benefits.
6. Recommend a CRM solution.
7. Design a customer feedback process.
8. Develop CRM performance indicators.
9. Create a CRM implementation plan.
10. Propose strategies to improve customer satisfaction.

Viva Questions

1. What is CRM?
2. Why is CRM important?
3. What are CRM modules?
4. What is customer retention?
5. What is customer satisfaction?
6. What is lead management?
7. What is automation?
8. What is customer loyalty?
9. What are CRM reports?
10. Name popular CRM systems.

6. IT Asset Management

Case Study

An organization owns 500 computers but lacks records of asset ownership and maintenance history.

Questions

1. Define IT Asset Management.
2. Explain the asset lifecycle.
3. Identify asset categories.
4. Analyze problems caused by poor asset tracking.
5. Evaluate asset inventory methods.
6. Recommend asset management tools.
7. Design an asset register.
8. Develop asset control procedures.
9. Create an asset auditing plan.
10. Propose a lifecycle management strategy.

Viva Questions

1. What is an IT asset?
2. What is an asset register?
3. What is asset depreciation?
4. Why conduct asset audits?
5. What is asset disposal?
6. What is software licensing?
7. What is inventory management?
8. What are asset tags?
9. What is lifecycle management?
10. Why is ITAM important?

7. Procurement Management

Case Study

A university plans to purchase 200 new computers for laboratories.

Questions

1. Define Procurement Management.
2. Explain procurement stages.
3. Identify procurement stakeholders.
4. Analyze procurement risks.
5. Evaluate supplier selection criteria.
6. Recommend procurement methods.
7. Design a procurement workflow.
8. Develop evaluation criteria for vendors.
9. Create a procurement policy.
10. Propose a transparent procurement process.

Viva Questions

1. What is procurement?
2. What is a tender?
3. What is RFQ?
4. What is RFP?
5. What is vendor evaluation?
6. What is contract management?
7. What is procurement risk?
8. What is ethical procurement?
9. What is supplier selection?
10. Why maintain procurement records?

8. Network Support Services

Case Study

Students complain about slow internet access and frequent Wi-Fi disconnections.

Questions

1. Define Network Support Services.
2. Explain network support responsibilities.
3. Identify possible causes of slow internet.
4. Analyze the network issues.
5. Evaluate troubleshooting methods.
6. Recommend solutions.
7. Design a monitoring strategy.
8. Develop preventive maintenance activities.
9. Create a support escalation procedure.
10. Propose a network improvement plan.

Viva Questions

1. What is network support?
2. What is bandwidth?
3. What is latency?
4. What is packet loss?
5. What is Wi-Fi interference?
6. What is network monitoring?
7. What is a switch?
8. What is a router?
9. What is DNS?
10. What tools are used in network troubleshooting?

9. Computer Maintenance

Case Study

A computer laboratory experiences frequent system failures and performance degradation.

Questions

1. Define Computer Maintenance.
2. Explain preventive and corrective maintenance.
3. Identify maintenance activities.
4. Analyze causes of failures.
5. Evaluate maintenance schedules.
6. Recommend maintenance tools.
7. Design a maintenance checklist.
8. Develop a maintenance policy.
9. Create a maintenance log template.
10. Propose a laboratory maintenance plan.

Viva Questions

1. What is preventive maintenance?
2. What is corrective maintenance?
3. Why clean hardware?
4. What causes overheating?
5. What is disk fragmentation?
6. What is antivirus software?
7. Why update operating systems?
8. What is backup?
9. What is hardware diagnosis?
10. What tools are used in maintenance?

10. Risk Management

Case Study

A university stores all student records in a single server without redundancy or backup.

Questions

1. Define Risk Management.
2. Explain risk management steps.
3. Identify risks in the case study.
4. Analyze the impact of each risk.
5. Evaluate existing controls.
6. Recommend mitigation strategies.
7. Design a risk assessment matrix.
8. Develop a risk register.
9. Create a risk response plan.
10. Propose a business continuity strategy.

Viva Questions

1. What is risk?
2. What is risk assessment?
3. What is risk mitigation?
4. What is likelihood?
5. What is impact?
6. What is residual risk?
7. What is risk register?
8. What is business continuity?
9. What is disaster recovery?
10. Why manage risks?

11. Change Management

Case Study

The IT department plans to upgrade the university's ERP system affecting all departments.

Questions

1. Define Change Management.
2. Explain the change lifecycle.
3. Identify stakeholders.
4. Analyze potential impacts.
5. Evaluate risks of the upgrade.
6. Recommend change controls.
7. Design a change approval process.
8. Develop a rollback plan.
9. Create a communication strategy.
10. Propose a successful implementation plan.

Viva Questions

1. What is change management?
2. What is a change request?
3. What is CAB?
4. What is rollback?
5. What is change approval?
6. What is change risk?
7. What is emergency change?
8. What is standard change?
9. Why document changes?
10. How measure change success?

12. Backup & Recovery

Case Study

A ransomware attack encrypts all files on the institute's file server.

Questions

1. Define backup and recovery.
2. Explain backup types.
3. Identify weaknesses in current practices.
4. Analyze ransomware impacts.
5. Evaluate backup strategies.
6. Recommend recovery procedures.
7. Design a backup schedule.
8. Develop a disaster recovery plan.
9. Create a recovery testing process.
10. Propose a backup policy.

Viva Questions

1. What is backup?
2. What is recovery?
3. What is full backup?
4. What is incremental backup?
5. What is differential backup?
6. What is RPO?
7. What is RTO?
8. Why test backups?
9. What is disaster recovery?
10. What is ransomware?

13. Service Desk Operations

Case Study

An organization receives over 500 support requests daily through phone, email, and web portal.

Questions

1. Define Service Desk.
2. Explain Service Desk functions.
3. Differentiate Service Desk and Help Desk.
4. Analyze operational challenges.
5. Evaluate ticket management approaches.
6. Recommend improvements.
7. Design a service request process.
8. Develop Service Desk KPIs.
9. Create staffing recommendations.
10. Propose an operational improvement plan.

Viva Questions

1. What is Service Desk?
2. What is ticket management?
3. What is escalation?
4. What is self-service?
5. What are KPIs?
6. What is SLA?
7. What is customer satisfaction?
8. What is incident management?
9. What is request fulfillment?
10. What is knowledge management?

14. Troubleshooting Process

Case Study

A lecturer cannot access the university LMS despite having internet access.

Questions

1. Define troubleshooting.
2. Explain troubleshooting methodology.
3. Identify possible causes.
4. Analyze the issue systematically.
5. Evaluate alternative solutions.
6. Recommend corrective actions.
7. Design a troubleshooting flowchart.
8. Develop diagnostic procedures.
9. Create a troubleshooting report.
10. Propose preventive measures.

Viva Questions

1. What is troubleshooting?
2. Why gather information first?
3. What is root cause analysis?
4. What is diagnosis?
5. What is verification?
6. What is documentation?
7. What is escalation?
8. What tools assist troubleshooting?
9. Why test solutions?
10. What is preventive maintenance?

15. Vendor Management

Case Study

A university relies on multiple suppliers for hardware, software, and networking services.

Questions

1. Define Vendor Management.
2. Explain vendor lifecycle stages.
3. Identify vendor selection criteria.
4. Analyze vendor-related risks.
5. Evaluate current vendor performance.
6. Recommend improvement measures.
7. Design a vendor evaluation framework.
8. Develop a vendor scorecard.
9. Create a vendor monitoring process.
10. Propose a vendor relationship strategy.

Viva Questions

1. What is a vendor?
2. What is vendor management?
3. What is a contract?
4. What is supplier evaluation?
5. What is SLA?
6. What is procurement?
7. What is vendor risk?
8. What is compliance?
9. What is performance monitoring?
10. Why maintain vendor relationships?

16. Documentation & Reporting

Case Study

An IT department resolves hundreds of issues each month but lacks proper documentation and reporting.

Questions

1. Define IT documentation.
2. Explain the importance of documentation.
3. Identify types of IT reports.
4. Analyze consequences of poor documentation.
5. Evaluate documentation standards.
6. Recommend reporting methods.
7. Design a documentation template.
8. Develop a reporting framework.
9. Create monthly IT service reports.
10. Propose a documentation improvement strategy.

Viva Questions

1. What is documentation?
2. Why document procedures?
3. What is a knowledge base?
4. What is an incident report?
5. What is a maintenance report?
6. What is report accuracy?
7. What is version control?
8. Why use templates?
9. What is audit evidence?
10. Why is reporting important?

Suggested Assessment Scheme

Component	Marks
Case Study Analysis	20
Questions 1–4 (Knowledge & Understanding)	20
Questions 5–7 (Application & Analysis)	25
Questions 8–10 (Evaluation & Creation)	25
Report Presentation & Formatting	10
Total	100

Viva Marking Scheme (20 Marks)

Criteria	Marks
Subject Knowledge	5
Application to Case Study	5
Critical Thinking	4
Communication Skills	3
Confidence & Professionalism	3
Total	20